

Scientific Writing

Anton Feenstra 2016
(slides Sanne Abeln)

Writing: understand your audience

- For any piece you are writing (thesis, exam question, poster, presentation, scientific paper, lecture) it is very important to keep your audience in mind.
- For now: write for a fellow MSc / BSc student (and not for your teachers!)

Explain, explain and teach

- The main aim of writing down what you have done, is explaining it to others. It follows that you need to:
 - be complete
 - be precise
 - be concise
 - be clear
 - make it interesting

Be complete

- Science is based on the concept of reproducible experiments:
 - someone should be able to reproduce your experiment based on your paper

Be precise

- Science, and specifically Bioinformatics and Systems Biology, are quantitative.
- Be precise (and correct) in your writing, and give quantities and references for statements.

Example:

... nowadays loads of sequences are present in sequence databases...

or

... in 2015 Uniprot[1] contained 630 million sequences...

Be concise

Concise: “giving a lot of information clearly and in a few words; brief but comprehensive.”

- Once you have written everything down in a complete and precise manner, go back over your text to make it clearer and more compact.
Cut every word that is not strictly necessary.
→ Remove all unnecessary words.

Be clear

- Highlight the most important parts
- Use a clear structure; create a story line
- In an average paper you can have one main point, and about five additional points – choose your points carefully – highlight them

Make it interesting

- Write a story
 - This is the story about answering your research question
 - Use abstract to set your story line
- Put in something of yourself:
 - Highlight a detail that you find particularly interesting
 - Make your reader aware of some interesting (side) result.
 - Summarize previous work, using your own interpretation
 - Note you can only do this a few times (2-3) in paper
- Use a **thesaurus**
 - but remain realistic:
 - “My canine decamp to the esplanade present.”

Some other ‘Rules’

“Use active voice”

– voices:

- The experiment was performed ... (passive)
- We performed an experiment ... (active)

– but:

exclusive use of the active voice is ‘American English’ style.

- Papers written solely in active voice quickly read like: ‘we did this, we did that, we are great’

Some other ‘Rules’

“Use present tense”

- Tenses:
 - “we observe that ...” (present simple)
 - “we are observing that ” (present continuous – generally, to be avoided)
 - “we observed that ...” (past simple)
 - “we have observed that” (present perfect)
 - “we had observed that” (past perfect –generally, to be avoided)
- Many papers are written in the past simple or present simple
- Make sure your tense is consistent.

Structure of a Scientific Paper

General structure:

- Abstract
- Introduction
- Materials and methods
- Results or Results and Discussion
- Discussion or Conclusion or Conclusions
- References

Note, specific fields and specific journals have their own structure conventions

Structure of a paper

Most importantly

- In all sections the connection to your **research question** should be clear and explicit
- You should be able to read the text without the headings
 - However, many people may first read the abstract and subsequently read the results section or look at Figures.

Abstract

- Use this to set out the story line (around a research question) for your paper
- Be concise, but complete, and try to aim it at a wide audience

Typical Structure:

- Motivation & research question
- Methods & results
- Conclusion & potential impact

For greater impact:

- make sure to connect it to current hot topics
 - any claim needs to be justified in the full text!

Introduction

In general:

- Aim: Introduce the research question
- Start at a more generic level, and slowly zoom in, and give context on the specific question (“funnel shaped introduction”)



https://commons.wikimedia.org/wiki/File:Kitchen_Funnel.jpg

Report:

- Include references to previous studies from literature
- Explain background of the (PSI-)BLAST and your database
- State your research question

Materials and methods

- Aim: allow people to reproduce the experiments you performed to answer your research question
- Be precise and provide sufficient technical detail (e.g. equations, parameter settings, pseudocode)
- Report:
 - Include a flow chart

Results and discussion

- Aim: show the evidence to answer your research question, and explain and interpret these results
- Include Figures and Tables to summarize evidence
- make sure to describe these results also in main text.

Results

Recipe for typical paragraph:

- in order to test
- we performed analysis
- Figure ... shows ...
- This explains / this is surprising / as expected

Sometimes iterating the last two points

Results and discussion

Report:

- Include your ROC plot and its interpretation.
- Answer your research question
- Compare your results to previous results from literature
- Discuss the (dis)advantages of using your chosen database as a gold standard for benchmarking homologs found with (PSI-)BLAST compared to the other two databases.

Discussion and conclusion

- Discuss any issues that may affect the results
- Clearly state what research question you have answered in this paper
- Explain what impact your research has on future research

Figures and Tables

Report:

- Explain all axes, labels, lines points, in the caption of your figure/table.
- Refer to each figure/table in the main text, and explain in the main text what can be seen from the figure/table.

General problems (also in Drafts):

- Plagiarism! (or, sloppy copy)
- Where to start introduction?
- Research Question!
- Conceptual misunderstanding
- Sloppy wording

Plagiarism – Be aware!

- Don't copy: read and reformulate.
- No entire sentences, figures or tables from other works – if you need to do so, use quotes and refer to the work.
- Always back up statements with references to published work (preferably not wikipedia, its not a primary source).
- The difference between getting inspired and plagiarism
 - There are now very serious consequences, rules are strictly applied.
 - Examination Board Fraud Committee

Citing vs. Copying

- “BLAST is a common alignment tool that compares nucleotide or protein sequences to a sequence database and calculates the statistical significance of putative homology.”
- <https://blast.ncbi.nlm.nih.gov/Blast.cgi>:
“BLAST finds regions of similarity between biological sequences. The program compares nucleotide or protein sequences to sequence databases and calculates the statistical significance.”
- “By aligning the sequences based on similarity, BLAST then gives a statistical significance of the matches for how likely they are to be homologs.”

Citing vs. Copying

- Recent advances in high-throughput sequencing methods have revolutionized the study of molecular biology and genomics and are used to study many biological questions (Soon et al., 2013).
- *Soon et al: “Advances in genome sequencing have progressed at a rapid pace, with increased throughput accompanied by plunging costs. But these advances go far beyond faster and cheaper. High-throughput sequencing technologies are now routinely being applied to a wide range of important topics in biology and medicine, often allowing researchers to address important biological questions that were not possible before.”*

General problems in Drafts:

- Where to start introduction?
 - Aim at general knowledge fellow MSc student (before course)
 - possible starting points are: the importance of BLAST or the importance of homology [both need to be introduced]

How to start?

- A critical step in identifying the function of a newly sequenced protein is employing databases to find proteins that contain sufficient similarities in function or sequence. These similar proteins could be homologs, meaning that they have a common ancestor.
- In a time where biological data is becoming increasingly available, rapid innovation and progress of sequence technologies is needed. Furthermore, techniques comparing genes and their products, such as putative homology analysis, could help provide information about unknown proteins.
- Protein homologue detection is central to bioinformatics. Several different methods for homologue detection are available (pairwise, profile-sequence, profile-profile) but the standards for evaluating their individual reliability leave a lot to be desired.
- Protein structure and function and often even nucleotide sequence are preserved so well between organisms that genes and proteins that share a common ancestor can be identified [4]. These genes and proteins are called homologs.

How to start? (2)

The increase in the amount of annotated DNA and protein sequences available and their structured storage in databases gives an opportunity to infer functional properties of newly sequenced genes or proteins by detecting homology. Several methods like BLAST or PSI-BLAST try to use this public knowledge to link sequence and functional similarities. However, their results are not always accurate and an assessment of their performance is required.

A challenge in bioinformatics is protein homology detection (Reid et al., 2007). Protein homology detection identifies homologs among proteins that share the existence of a common ancestor during evolution [Q2.1]. To detect close homologs, sequence similarity is effective and reliable due the fact that sequences that share significant similarity can be homologous.

How to start? (3)

- BLAST identifies regions of similarity between biological sequences [1]. When two protein sequences are similar, they are potentially homologous.
- Recognition of homologous sequences is a key aspect to learn about a new sequence.
- Homology detection is an important tool to gain additional information of for example protein structure or function by sequence analysis.
- A critical step in identifying the function of a newly sequenced protein is employing databases to find proteins that contain sufficient similarities in function or sequence. These similar proteins could be homologs, meaning that they have a common ancestor.
- Sequence alignment is often the first step in the analysis of a newly found protein. Sequences that are similar could be homologs, which means that they share a common ancestor. Identifying homologs is useful because it implies a common structure and function of the unknown protein.

How to start? (4)

- Two genes or proteins are homologous if they share a common evolutionary origin [5]. This information is valuable in the classification of newly discovered genes, as homology with a known gene can serve as a predictor of structure and function. The Basic Local Alignment Search Tool (BLAST) is a computer program designed to find putative homologs by aligning a query sequence to a sequence database. During a BLAST run, the similarity between the sequences is calculated, and when the sequence similarity is high enough, the query and database sequence are predicted to be homologous [6]. The structure and function of the known gene or protein can then be transferred to its putative homolog.

General problems in Drafts:

- Research question !!

(missing, or not well formulated)

Research Question

- Is SCOP a reliable database to benchmark two alignment tools BLAST and PSI-BLAST?
- In this project we aim to benchmark the protein homology search tool of BLAST and PSI-BLAST using SCOP as a golden standard for homology.
- The main objective of this project is to benchmark (PSI-)BLAST against the SCOP database, which classifies protein domains based on their structure, using a local database of 212 proteins, and to identify which SCOP classifier generates more true homologs: family or superfamily.

Research Question (2)

- this article aims to benchmark the performance of BLAST and PSI-BLAST when assessing a GO gold standard datasetthis article aims to benchmark the performance of BLAST and PSI-BLAST when assessing a GO gold standard dataset
- the present report questions whether or not Pfam database could serve as a gold standard for predicting protein sequence homology.
- How well do BLAST and PSI-BLAST perform in detecting putative homology judged by their sensitivity, and error? And how do they compare?

Research Question (3)

- The goal of our research was to detect homology in a set of proteins using the algorithms and tools provided in BLAST and PSI-BLAST.
- The objective of this study is to evaluate the performance BLAST compared to PSI-BLAST using a predefined library of protein Ids.
- The objective of this report is to compare the two methods(BLAST - (PSI-BLAST) and mention which is better.

General problems in Drafts:

- Conceptual misunderstanding of terms, e.g.:
 - Sometimes there is real misunderstanding
 - Sometimes it is sloppy wording
- Both are unacceptable

What's the problem?

"PSI-BLAST generates more bad hits than BLAST"

"SCOP database is bigger than the others"

What's the problem?

"The e-value corresponds to the expected number of hits that are false positive"

"Because BLAST is more strict than PSI-BLAST, it does not have many true positives"

"In BLAST, sequences with low E-values are said to be homologous."

What's the problem?

"The AUC represents the probability that a true positive scores higher than a true negative."

General problems in Drafts:

- Conceptual misunderstanding of terms, e.g.:
 - benchmark gold standard vs database
 - homology vs BLAST output
 - NA and ambiguous terms
 - e-value is probability
 - “running pfam”
 - ROC curve is a parametric curve of an e-value threshold

TASK

- Write a 200 word abstract for your project

Abstract

- Use this to set out the story line (around a research question) for your paper
- Be concise, but complete, and try to aim it at a wide audience

Structure [200 words max]:

- Motivation & research question
- Methods & results
- Conclusion & potential impact

For greater impact:

- make sure to connect it to current hot topics
 - this of course needs to be backed-up by the paper

Criteria for feedback

Excellent, Good, Sufficient, Insufficient:

- Is the research question clear?
- Is the story line clear?
- Is it clear how the research was performed?
- Is it clear why this research question is interesting to answer?

Give two positive comments, and two critical comments

Suggest 10 words that can be removed.

Homology detection is an important tool in bioinformatics. The aim of this report is to research whether (PSI-)BLAST predictions of homology are corroborated by the Pfam database. We used (PSI-)BLAST to predict homology between a Uniprot_ID list of 212 proteins and checked this with the Pfam database.

The Receiver Operator Characteristics (ROC) plots of BLAST and PSI-BLAST have respective Area Under Curves (AUC)'s of 0.924 (BLAST) and 0.925 (PSI-BLAST). These results suggest that both BLAST and PSI-BLAST are appropriate tools for predicting homology between proteins. The usage of Pfam as a gold standard database is something arguable when comparing it with Gene Ontology (GO) and Structural Classification Of Proteins (SCOP) databases. Future studies might focus on the usage of Pfam as a gold standard database.

Is the research question clear?

Is the story line clear?

Is it clear how the research was performed?

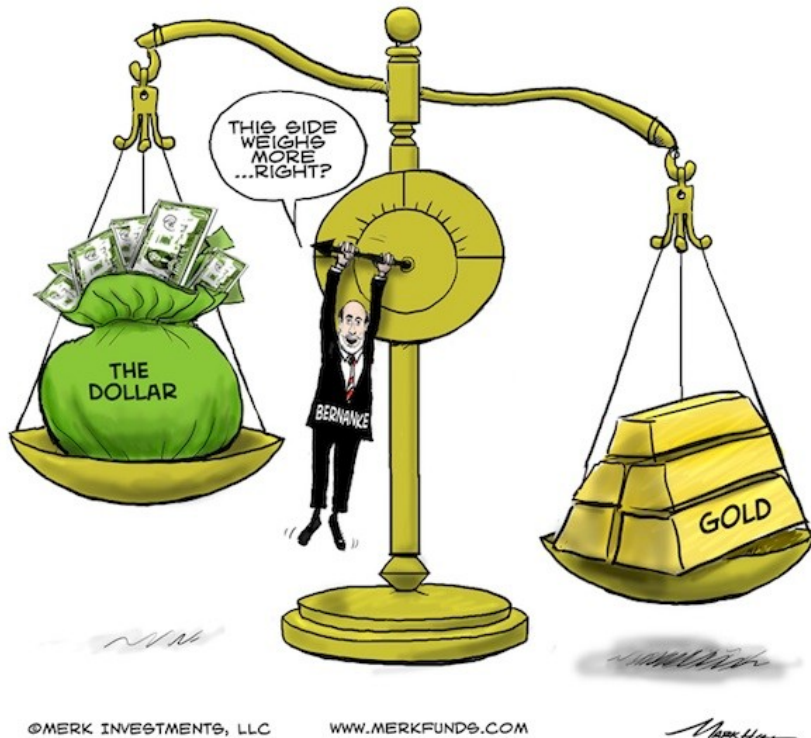
Is it clear why this research question is interesting to answer?

Homology detection is an important tool in bioinformatics. The aim of this report is to research whether (PSI-)BLAST predictions of homology are corroborated by the Pfam database. We used (PSI-)BLAST to predict homology between a Uniprot_ID list of 212 proteins and checked this with the Pfam database.

The Receiver Operator Characteristics (ROC) plots of BLAST and PSI-BLAST have respective Area Under Curves (AUC)'s of 0.924 (BLAST) and 0.925 (PSI-BLAST). These results suggest that both BLAST and PSI-BLAST are appropriate tools for predicting homology between proteins. The usage of Pfam as a gold standard database is something arguable when comparing it with Gene Ontology (GO) and Structural Classification Of Proteins (SCOP) databases. Future studies might focus on the usage of Pfam as a gold standard database.

Give two positive comments, and two critical comments
Suggest 10 words that can be removed.

Gold standard – Golden standard



See also “The gold standard: not a golden standard” by Jurgen Claassen
www.ncbi.nlm.nih.gov/pmc/articles/PMC557893/